

Vamsi Krishna Pamulapati

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AI Engineer and Software Developer building scalable backend systems, API-driven architectures, and intelligent automation workflows; experienced at Comcast developing enterprise distributed applications and workflow automation systems using TIBCO Business Works while pursuing an M.S. in Data Science & Artificial Intelligence.

EXPERIENCE

Comcast | Development Engineer 1

July 2024 – July 2025

Tibco Developer

Chennai, India

- Designed and developed 15+ enterprise integration services enabling real-time data exchange across Salesforce, SAP, Hermes, and internal business platforms using event-driven & service-oriented architectures.
- Developed and maintained REST and SOAP APIs using TIBCO Business Works to process enterprise business transactions and support seamless interoperability across CRM and ERP systems.
- Automated business-critical processes and orchestration pipelines, reducing manual intervention by 30% while increasing operational reliability and process consistency.
- Collaborated with cross-functional engineering teams to troubleshoot production incidents, perform root-cause analysis, and maintain reliability of enterprise-scale integration services.
- Implemented scalable workflow orchestration patterns as part of integration modernization initiatives, improving maintainability, extensibility, and operational efficiency of enterprise services.
- Implemented monitoring, validation, and error-handling mechanisms to improve service resilience and ensure reliable data movement across enterprise systems.

Comcast | Development Engineer 1

Jan 2024 – Jun 2024

Tibco Developer - Intern

Chennai, India

- Developed and supported enterprise integration services using TIBCO Business Works, enabling reliable communication between CRM platforms and internal business systems.
- Assisted in designing and implementing API-driven workflows and middleware solutions for enterprise system integration and business process automation.
- Troubleshot production incidents performed root-cause analysis and contributed to improving system stability and service reliability.
- Collaborated with senior engineers on integration architecture enhancements, workflow automation initiatives, and platform modernization efforts.
- Conducted technical research and feasibility analysis to support process optimization and enterprise integration improvements.
- Coordinated cross-functional efforts with QA and engineering teams to streamline validation workflows, improving delivery efficiency across validation and release cycles by 15%.

Usurp HR Tech Solutions.

Sept 2023 – Dec 2023

Web Developer - Freelance

India (Remote)

- Developed responsive web applications using HTML, CSS, JavaScript, and Django, delivering user-facing systems with secure authentication and scalable backend architecture.
- Built backend functionality in Django, including authentication workflows and application security controls.
- Applied user-centered design (UCD), design thinking, and agile methodologies to improve user experience and optimize interaction flows, increasing user engagement by 35%.
- Collaborated with stakeholders to translate business requirements into production-ready web solutions and ensure timely feature delivery.

PROJECTS

Gen AI for Ad Recommendation Using Multimodal Historical Behaviour Data **09/2025 – Present**

- Built a multimodal recommendation system using transformer-based architectures that integrates user interaction histories, textual metadata, and visual embeddings to generate personalized ad recommendations. Implemented using PyTorch, Hugging Face Transformers, and Python.
- Developed transformer-based ranking and recommendation pipelines to model user behavior and improve engagement prediction from heterogeneous data sources.
- Integrated fairness-aware evaluation metrics and Responsible AI frameworks to assess bias, improve recommendation transparency, and ensure equitable model outputs.
- Ongoing research toward submission to RecSys 2026, focusing on generative recommendation modeling and multimodal user representation learning.

Copernico Payment Processing System. **03/2025 – 07/2025**

- Contributed to the development of an intelligent payment processing platform supporting transaction routing and operational monitoring.
- Applied analytics-driven approaches for transaction validation, anomaly identification, and payment workflow optimization.
- Supported integration with multiple payment gateways to enhance processing flexibility and reliability.

AI-Driven Billing Assistant Using Mobile Vision and Web Development. **03/2024 – 10/2024**

- Developed an AI-powered mobile billing assistant integrating computer vision, QR-code workflows, and web technologies to automate invoice generation and customer self-service transactions.
- Focused on improving customer convenience through automation and mobile-first interactions.
- Integrated image capture capabilities to support billing verification and digital transaction processing.

Email Spam Classification Using Machine Learning. **08/2023 – 03/2024**

- Built an email classification model using ML techniques to distinguish spam from legitimate messages.
- Performed data preprocessing, feature engineering, model training, and evaluation using Python ML libraries.
- Applied classification methodologies to improve email filtering and security workflows.

PUBLICATIONS

Titanic Survival Analysis Using Logistic Regression.

- Developed a binary classification model using logistic regression to predict passenger survival outcomes from demographic and travel-related features.
- Investigated survival probability relationships based on passenger demographics & travel characteristics.

SKILLS

- **Technical Skills:** C, Java, Python, MySQL, No-SQL, DSA, Excel, Shell & Unix Scripting
- **Tools:** Git, GitHub, Docker, Jenkins, Tibco BW, Flowable BPM, n8n
- **Technologies:** AI&ML, Data Science, Deep Learning, NLP, Cloud Computing, Web Development
- **Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, Django
- **Soft Skills:** Teamwork, Leadership, Problem-solving, Agile & Scrum

EDUCATION

University of Central Missouri, Warrensburg
M.S in Data Science and AI

Aug 2025 – May 2027

IBM Artificial Intelligence And Machine Learning Graduate
Professional Certificate

Aug 2020 – May 2024